Java Swing Tutorial

Java Swing tutorial is a part of Java Foundation Classes (JFC) that is used to create window-based applications. It is built on the top of AWT (Abstract Windowing Toolkit) API and entirely written in java.

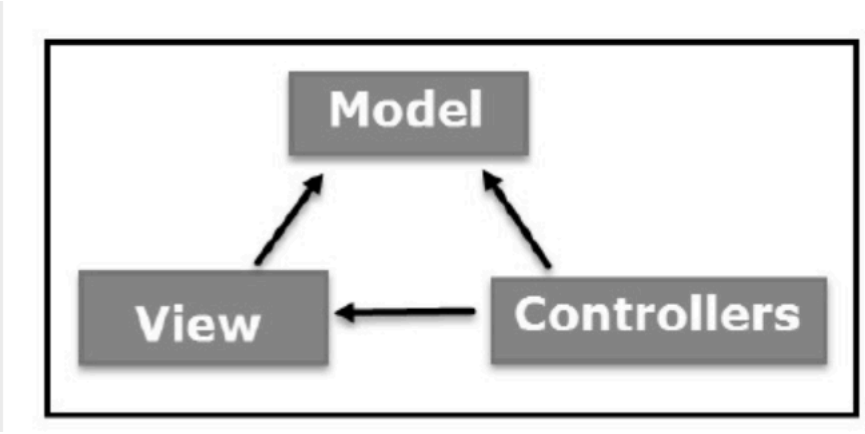
Unlike AWT, Java Swing provides platform-independent and lightweight components.

The javax.swing package provides classes for java swing API such as JButton, JTextField, JTextArea, JRadioButton, JCheckbox, JMenu, JColorChooser etc.

Difference between AWT and Swing

No.	Java AWT	Java Swing
1)	AWT components are **platform-dependent**.	Java swing components are **platform-independent**.
2)	AWT components are **heavyweight**.	Swing components are **lightweight**.
3)	AWT **doesn't support pluggable look and feel**.	Swing **supports pluggable look and feel**.
4)	AWT provides **less components** than Swing.	Swing provides **more powerful components** such as tables, lists, scrollpanes, colorchooser, tabbedpane etc.
5)	AWT **doesn't follows MVC**(Model View	Swing **follows MVC**.

The **Model-View-Controller (MVC)** is an architectural pattern that separates an application into three main logical components: the **model**, the view, and the controller. Each of these components are built to handle specific development aspects of an application. MVC is one of the most frequently used industry-standard web development framework to create scalable and extensible projects.



Model

The Model component corresponds to all the data-related logic that the user works with.

View

The View component is used for all the UI logic of the application. For example, the Customer view will include all the UI components such as text boxes, dropdowns, etc. that the final user interacts with.

Controller

Controllers act as an interface between Model and View components to process all the business logic and incoming requests, manipulate data using the Model component and interact with the Views to render the final output.

What is component

A component is **an object having a graphical representation that can be displayed on the screen and that can interact with the user**. Examples of components are the buttons, checkboxes, and scrollbars of a typical graphical user interface.

What is container

Containers are **the interface between a component and the low-level, platform-specific functionality that supports the component**.

What is servlet

- Servlet is a technology which is used to create a web application.
- Servlet is an API that provides many interfaces and classes including documentation.
- Servlet is an interface that must be implemented for creating any Servlet.
- Servlet is a class that extends the capabilities of the servers and responds to the incoming requests. It can respond to any requests.

- Servlet is a web component that is deployed on the server to create a dynamic web page

What is Applet?

An applet is a Java program that can be embedded into a web page. It runs inside the web browser and works at client side. An applet is embedded in an HTML page using the APPLET or OBJECT tag and hosted on a web server.

Applets are used to make the website more dynamic and entertaining.

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Java AWT Tutorial

Java AWT (Abstract Window Toolkit) is *an API to develop Graphical User Interface (GUI) or windows-based applications* in Java.

Java AWT components are platform-dependent i.e. components are displayed according to the view of operating system. AWT is heavy weight i.e. its components are using the resources of underlying operating system (OS).

The java.awt package provides classes for AWT API such as TextField, Label, TextArea, RadioButton, CheckBox, Choice, List etc.

What is JFrame

JFrame is a top-level container that provides a window on the screen. A frame is actually a base window on which other components rely, namely the menu bar, panels, labels, text fields, buttons, etc. Almost every other Swing application starts with the JFrame window.

What is Jpanel

JPanel, a part of the Java Swing package, is a container that can store a group of components. The main task of JPanel is to organize components, various layouts can be set in JPanel which provide better organization of components; however, it does not have a title bar.

What is JTextPane

JTextPane is a subclass of JEditorPane class. JTextPane is used for styled document with embedded images and components. It is text component that can be marked up with attributes that are represented graphically. JTextPane uses a DefaultStyledDocument as its default model.

What is JButton

The JButton class is **used to create a labeled button that has platform independent implementation**. The application result in some action when the button is pushed. It inherits AbstractButton class.

What is interface

An **interface in Java** is a blueprint of a class. It has static constants and abstract methods. The interface in Java is *a mechanism to achieve abstraction*. There can be only abstract methods in the Java interface, not method body. It is used to achieve abstraction and multiple inheritance in Java. In other words, you can say that interfaces can have abstract methods and variables. It cannot have a method body.

A **Thread** is a very light-weighted process, or we can say the smallest part of the process that allows a program to operate more efficiently by running multiple tasks simultaneously.

The **@Override** annotation indicates that the child class method is over-writing its base class method. The @Override annotation can be useful for two reasons. It extracts a warning from the compiler if the annotated method doesn't actually override anything. It can improve the readability of the source code

SerialVersionUID is a unique identifier for each class, JVM uses it to compare the versions of the class ensuring that the same class was used during Serialization is loaded during Deserialization. Specifying one gives more control, though JVM does generate one if you don't specify**.**

The printStackTrace() method in Java is a tool used to handle exceptions and errors. It is a method of Java's throwable class which prints the throwable along with other details like the line number and class name where the exception occurred. printStackTrace() is very useful in diagnosing exceptions.

The Java FlowLayout class is used to arrange the components in a line, one after another (in a flow). It is the default layout of the applet or panel.

The setLayout(...) method allows you to set the layout of the container, often a JPanel, to say FlowLayout, BorderLayout, GridLayout, null layout, or whatever layout desired. The layout manager helps lay out the components held by this container.

As in the above code, the **@SuppressWarnings** annotation is annotated to the class as @SuppressWarnings("deprecation"), which means now all methods inside the class are also applied the warning deprecation to be to suppress, and now the compiler will not issue a warning in both methods line

What is **actionPerformed** in java

An action event occurs, whenever an action is performed by the user. Examples: When the user clicks a button, chooses a menu item, presses Enter in a text field. The result is that an actionPerformed message is sent to all action listeners that are registered on the relevant component.

what is **ActionEvent** in java

public class ActionEvent extends AWTEvent. A semantic event which indicates that a component-defined action occurred. This high-level event is generated by a component (such as a Button) when the component-specific action occurs (such as being pressed)